Final Paper – AI Perception

DACSS 790Q - Advanced Quantitative Methods

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Abstract

In everyday settings, individuals' interaction with AI (Artificial Intelligence) appears to evoke both excitement and concern. Some studies show that familiarity with AI correlates with positive perceptions while some other studies report that the same exposure can raise worries. This study investigates whether the frequency of interaction with artificial intelligence influences individuals' perceptions of AI. Specifically, it examines whether regular AI exposure correlates with more positive or neutral or negative concerns. Using data from a nationally representative sample, I leveraged an Ordered logit Model, Multinomial Model and Exploratory Factor Analysis (EFA) to identify underlying dimensions of AI perception. The findings indicate that heightened AI interaction generally correlates with more favorable perceptions, but this association is heavily influenced by demographic factors. Notably, the substantial effects in the multinomial model's "more excited" category are particularly noteworthy, demonstrating that frequent AI users exhibit a significantly higher tendency to express enthusiasm compared to neutral attitudes.

Introduction

The significance of researching whether enhanced interaction with artificial intelligence (AI) in daily life influences individuals' feelings of excitement or concern is paramount. As AI becomes increasingly integrated into both of our personal and professional lives, it immensely impacts emotional responses and societal dynamics. A comprehensive understanding of these emotional reactions is essential for designing human-centric AI systems and applications that enhance user experiences and guide governance policies to effectively address public apprehensions while simultaneously harnessing excitement.

In this paper, I will provide an overview of existing literature, outline the background of the research question, discuss the influence of the independent variables on the dependent variables, describe the research design setup, and present the analysis and results

Literature Review, Research Question & Hypothesis

Literature Review

AI is integrated in several aspects of our lives and has led to varied debates on its implications on our society. Per Bozkurt and Gursoy (2023), while many people believe that AI will simplify our lives, increase output in terms of efficiency, there are concerns about increased unemployment and social inequalities. In addition, individual perceptions are influenced by demographic factors such as age, gender and prior experience with AI. These findings highlight a significant confusion and uncertainty regarding the role and perception of artificial intelligence in society.

Kim, Ham, and Lee (2024) examined how student's attitudes toward AI and skills affect their interactions with AI during creative tasks. Their study found that students approach to collaborating with AI differ based on their openness to AI and their drawing abilities. Positive attitudes and higher skills led to deeper and more effective interactions, while negative attitudes or lower skills resulted in different patterns. These findings suggest that attitudes and user characteristics shape perceptions of AI.

Understanding public perception is essential for responsible research and innovation. The Public Attitudes to Data and AI Tracker Survey by Department for Science, Innovation and Technology – Government of United Kingdom (2024) reveals that despite growing public trust and improved optimism about data practices, persistent concerns remain about data security, commodification, lack of choice in data sharing, and surveillance. However, the public increasingly recognizes the beneficial role data and AI can play in designing products and services.

In the healthcare sector, professionals' perceptions of artificial intelligence (AI) are being assessed using tools such as the Shinners Artificial Intelligence Perception (SHAIP) questionnaire. This instrument identified two pivotal factors influencing healthcare professionals' viewpoints: the "professional impact of AI" and the "preparation for AI." Research via Shinners et. al (2022) suggests that the "use of AI" significantly impacts perceptions across both factors.

Per Brauner et. al (2023), AI becomes increasingly omnipresent, understanding the complex perception patterns is vital for addressing ethical concerns, guiding policy development, and designing human-centered AI systems that align with individual and societal needs.

Research Question & Hypothesis

With inspiration based on the above literature review, the research in this paper focuses on whether increased interaction with AI in our daily lives influences perception of AI. I will test the hypothesis: People who interact more frequently with artificial intelligence will report higher levels of excitement (or concern) about its increased use in daily life than those who interact less frequently.

Independent Variables influence on the Dependent Variable

The dependent variable in this paper is AI perception. The following shows how a few of the independent variables might influence the dependent variable:

Frequency of Internet Usage

Per Taib et. al (2019) academicians with higher internet competency reported more favorable views of digital tools, suggesting that frequent use enhances comfort with emerging technologies

Frequency of AI Usage
Marquis et. al (2024) show that professionals who regularly use AI for data analysis or decision-making report higher satisfaction and trust, as repeated interactions demystify AI's "black box" nature.

Perception of AI Assisted Robotic Surgery
Research by Torrent-Sellens et. al (202) shows that variables of a sociodemographic nature played an important predictive role. It was confirmed that the effect of experience on trust in robots for surgical interventions was greater among men, people between 40 and 54 years old, and those with higher educational levels.

Economic impact of AI
Briggs, Kodnani (2023) argue that significant uncertainty around the potential of generative AI, its ability to generate content that is indistinguishable from human-created output and to break down communication barriers between humans and machines reflects a major advancement with potentially large macroeconomic effects.

Comfort with AI in Healthcare

Per Li & Qin (2023), Higher comfort levels suggest greater trust in AI's reliability and safety, which should strongly correlate with feeling more excited and less concerned about its increasing integration into daily life.

The increased frequency of internet usage likely enhances exposure to information and discussions pertaining to artificial intelligence, thereby contributing to heightened awareness of AI.

Research Design

Sponsor(s)
This research paper does not have a sponsor

Data
The data was downloaded from the Pew Research Center.

American Trends Panel Wave 119
Field dates: Dec. 12 – Dec. 18, 2022
Survey Question Topics: AI and human enhancement

The target population for this survey was non-institutionalized persons ages 18 and older, living in the U.S., including Alaska and Hawaii. There are a total of 11,004 observations, each corresponding to a single respondent.

There are 4 columns that show related geographic areas:

F_METRO Metropolitan area indicator
F_CREGION Census region
F_CDIVISION Census division
F_USR_SELFID Self-identified Urban, Suburban, Rural (USR)

However, checking ICC (interclass correlations) shows that they are all below .1 indicating that there are negligible group level differences.

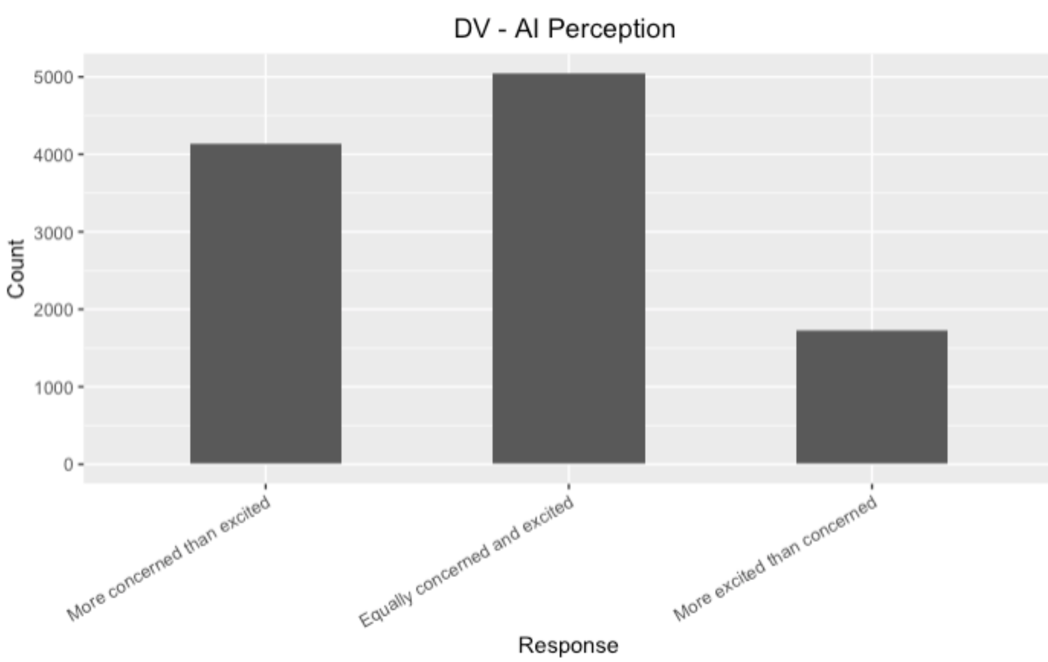
Dataset Link (requires a login)
https://www.pewresearch.org/dataset/american-trends-panel-wave-119/

Dependent Variable

The Dependent Variable (DV) is AI Perception based on the following categorical variable from the dataset:

CNCEXC_W119 - Overall, would you say the increased use of artificial intelligence (AI) in daily life makes you feel…
1 More excited than concerned
2 More concerned than excited
3 Equally concerned and excited
99 Refused

Distribution of the dependent variable after recoding, so that higher value is more positive



Primary Modeling Approach

The Primary model being utilized is the **Ordered logit** since the dependent variable is ordinal with 3 ordered categories

Independent Variables (IVs)

1. F_INTFREQ (Frequency of internet use)

This was asked to all survey participants. Choices included:

Almost constantly = 1, Several times a day = 2, About once a day = 3, Several times a week = 4, Less often = 5, Refused = 99

2. USEAI_W119 (Frequency of interaction with AI)

This was asked to all survey participants. Choices included:

Almost constantly = 1, Several times a day = 2, About once a day = 3, Several times a week = 4, Less often = 5, Refused = 99

3. SROBOT2_W119 (Robots with artificial intelligence (AI) are being developed that can be used to perform parts of surgery on their own. These robots are designed to make precise and steady movements during surgery. How much of an advance for medical care are surgical robots with artificial intelligence (AI) that can perform parts of surgery on their own?

This was asked on form 2 only, Choices included:

A major advance = 1, A minor advance = 2, Not an advance = 3, Not sure = 9, Refused = 99

4. AIEXT2_W119 (How much of an advance for weather forecasting is using artificial intelligence (AI) to predict when and where extreme weather is likely to occur?)

This was asked on form 2 only, about half of all respondents. Choices included:

A major advance = 1, A minor advance = 2, Not an advance = 3, Not sure = 9, Refused = 99

5. AIHCCOMF_W119 (Thinking about the use of artificial intelligence (AI) in health and medicine to do things like diagnose disease and recommend treatments, How would you feel if your health care provider relied on AI to do things like diagnose disease and recommend treatments?)

This was asked to all survey participants. Choices included:

1 = Very comfortable, 2 = Somewhat comfortable, 3 = Somewhat uncomfortable, 4 = Very uncomfortable, 99 = Refused

6. AIWRK2_c_W119 (Over the next 20 years, how much impact do you think the use of artificial intelligence (AI) in the workplace will have on the U.S. economy)

This was randomized among other similar questions but was asked to all respondents. Choices included:
A major impact = 1, A minor impact = 2, No impact = 3, Not sure = 9, Refused = 99

Control Variables

1. **F_GENDER (Gender)** – The gender could influence adoption and perception of AI. Due to risks, women might be more concerned than men
2. **F_AGECAT (Age Category)** – Younger folks seem to be early adopters and more excited to use AI, while older folks might be more skeptical about it
3. **F_EDUCCAT (Education Level)** – People with higher levels of education might be more familiar with technology and possibly grasp concepts within the field of AI better than those with lower levels of education. People with lower levels might also feel more conscious about impact of AI on their jobs.
4. **F_RACETHNMOD (Race-Ethnicity)** – I read a book last year "Unmasking AI" by Dr. Joy Buolamwini that showcased how racial bias is baked in to several AI algorithms and so people belonging to minority racial groups might feel more apprehensive towards AI.

Secondary Modeling Approach

The Secondary modeling approach being utilized is **Factor analysis/alpha calculation with new scale creation**

I need to create a new scale because combining multiple variables that measure a same construct provides better reliability than individual variables as seen in the study by Grassini, S. (2023) and hopefully the new scale reduces the impact of measurement error in any of the individual variables.

Analysis & Results

Approach 1 - Ordered Logit Model

Ordered Logit Model Results

	Dependent variable:
	AI Perception
Internet Usage - Several times a week	-0.036 (0.296)
Internet Usage - About once a day	-0.118 (0.279)
Internet Usage - Several times a day	0.025 (0.257)
Internet Usage - Almost constantly	0.091 (0.260)
AI Usage - Several times a week	0.290^{***} (0.082)
AI Usage - About once a day	0.390^{***} (0.088)
AI Usage - Several times a day	0.476^{***} (0.073)
AI Usage - Almost constantly	0.522^{***} (0.132)
AI surgery - Not an advance	-0.093 (0.141)
AI surgery - A minor advance	-0.051 (0.090)
AI surgery - A major advance	0.366^{***} (0.075)
AI weather - Not an advance	-0.686^{***} (0.153)
AI weather - A minor advance	-0.122 (0.085)
AI weather - A major advance	0.210^{***} (0.079)
AI healthcare - Somewhat uncomfortable	0.465^{***} (0.078)
AI healthcare - Somewhat comfortable	1.160^{***} (0.082)
AI healthcare - Very comfortable	1.935^{***} (0.131)
AI econ impact - Not an advance	0.418^{*} (0.224)
AI econ impact - A minor advance	0.172^{*} (0.097)
AI econ impact - A major advance	-0.132 (0.084)
Gender - Female	-0.303^{***} (0.057)
Gender - Other	-0.650^{**} (0.300)
Age category - 30-49	-0.168 (0.103)
Age category - 50-64	-0.316^{***} (0.108)
Age category - 65+	-0.358^{***} (0.113)
Education level - Some College	0.079 (0.083)
Education level - College graduate+	0.142^{*} (0.081)
Race - Black	0.174^{**} (0.086)
Race - Hispanic	0.301^{***} (0.086)
Race - Other	0.040 (0.152)
Race - Asian	0.584^{***} (0.150)
Observations	5,143
Note:	*p<0.1; **p<0.05; ***p<0.01

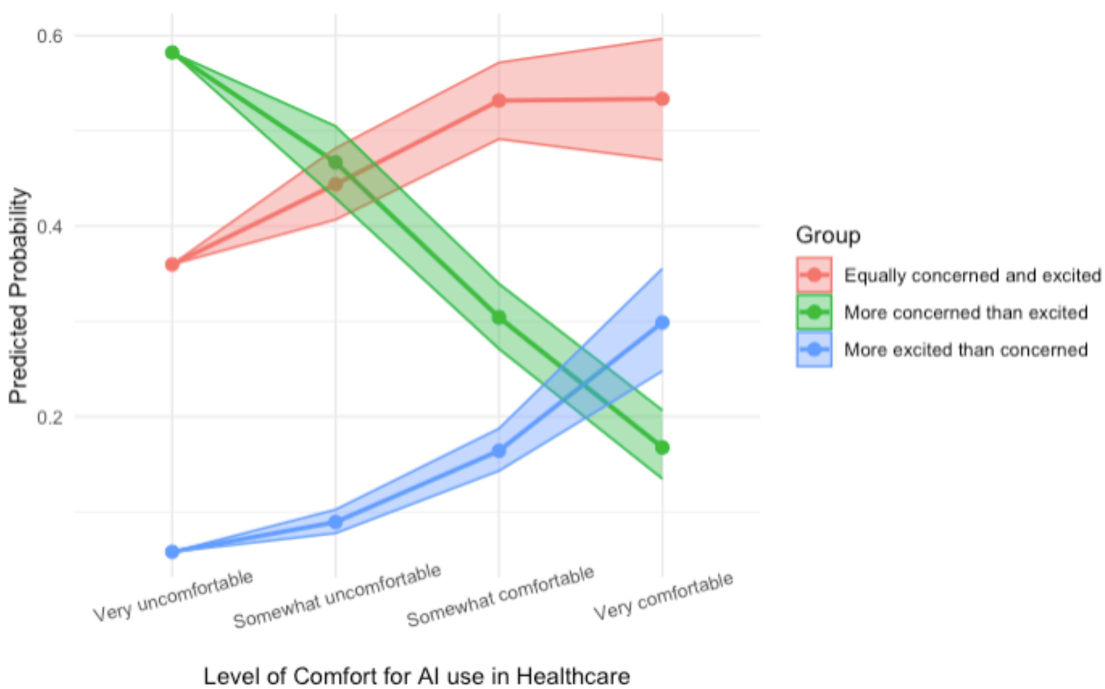
The results from running the ordered logit model above show that there's a clear relationship between AI usage frequency and positive AI perception. As the AI usage frequency increases from low usage (several times a week to almost constantly), the coefficient increases in addition to being statistically significant. This supports the hypothesis that increase in the usage of AI leads to high level of excitement.

Internet Usage as a whole shows no significant relationship with AI perception and so indicates that general internet use doesn't have any effect on AI perception.

More interestingly, Comfort with AI in Healthcare shows as the *Strongest Predictor*. The survey respondents that are “very comfortable” with AI being used in healthcare have approximately 6.92 times higher odds of being in the higher perception of AI category. The model shows that as people become more comfortable with AI in healthcare, their perception of AI shifts dramatically in a positive direction. The coefficients for comfort levels are large and highly significant

The demographic control variables show some interesting relationships:

1. Women and people of other genders have a very negative perception of AI when compared to Men (reference category)
2. As age increases, the perception of AI becomes more negative with individuals when compared to the 18 to 29 age group (reference category)
3. College graduates have a slightly more positive perception of AI when compared to individuals with a high school degree or less (reference category)
4. When compared to individuals that are White, Asians are associated with the highest positive perception of AI, followed by positive perceptions by Hispanics and Blacks



The predicted probability plot above shows varying probability patterns associated with AI perception based on level of comfort for AI use in healthcare. The plot indicates that individuals who are in the “More concerned than excited” group tend to maintain relatively consistent probability levels with a consistent downtrend indicating a negative relationship between their perception of AI versus their comfort of AI use in healthcare.

The other two groups show that as the individuals become more comfortable with AI use in healthcare, the predicted probability shifts more dramatically indicating a non-linear but positive relationship between perception of AI versus comfort of AI use in healthcare.

Brant Test for Parallel Regression Assumption

	X^2	df	probability
Omnibus	115.870	31	0.000
Internet_UsageSeveral times a week	0.068	1	0.794
Internet_UsageAbout once a day	0.903	1	0.342
Internet_UsageSeveral times a day	0.198	1	0.656
Internet_UsageAlmost constantly	0.760	1	0.383
AI_UsageSeveral times a week	0.124	1	0.725
AI_UsageAbout once a day	0.398	1	0.528
AI_UsageSeveral times a day	0.175	1	0.675
AI_UsageAlmost constantly	2.884	1	0.089
AI_surgeryNot an advance	7.276	1	0.007
AI_surgeryA minor advance	4.095	1	0.043
AI_surgeryA major advance	2.325	1	0.127
AI_weatherNot an advance	0.113	1	0.736
AI_weatherA minor advance	0.240	1	0.624
AI_weatherA major advance	0.705	1	0.401
AI_health_careSomewhat uncomfortable	0.998	1	0.318
AI_health_careSomewhat comfortable	1.168	1	0.280
AI_health_careVery comfortable	7.367	1	0.007
AI_econ_impactNot an advance	6.947	1	0.008
AI_econ_impactA minor advance	8.684	1	0.003
AI_econ_impactA major advance	16.142	1	0.000
GenderFemale	7.195	1	0.007
GenderOther	0.063	1	0.802
Age_category30-49	1.487	1	0.223
Age_category50-64	5.857	1	0.016
Age_category65+	1.697	1	0.193
Education_levelSome College	2.547	1	0.111
Education_levelCollege graduate+	0.744	1	0.388
RaceBlack	3.035	1	0.081
RaceHispanic	3.399	1	0.065
RaceOther	0.108	1	0.742
RaceAsian	0.507	1	0.477

The results of running the brant test (to test the parallel regression assumption) is shown above. The omnibus test is highly significant, indicating an overall violation.

Several other variables also have violations:

1. AI Surgery – Not an Advance
2. AI Healthcare – Very Comfortable
3. AI Economic Impact
4. Gender – Female
5. Age Category 50 to 64

The results of the brant test indicates that the variables mentioned above have different effects across the levels of AI Perception (as opposed to consistent effects that the

Ordered Logit model assumes). Due to the violations, a **Multinomial model** is then utilized that treats the DV (AI Perception) as an unordered category than ordinal.

Approach 1 – continued with Multinomial Model

Multinomial Model Results

	Dependent variable:	
	Equally concerned and excited	More excited than concerned
	(1)	(2)
(Intercept)	0.01 (0.32)	0.19 (0.65)
Internet Usage - Several times a week	-0.21 (0.30)	0.32 (0.61)
Internet Usage - About once a day	0.06 (0.28)	0.38 (0.58)
Internet Usage - Several times a day	-0.02 (0.28)	0.55 (0.58)
Internet Usage - Almost constantly	0.24^{**} (0.10)	0.46^{***} (0.14)
AI Usage - Several times a week	0.31^{***} (0.10)	0.65^{***} (0.15)
AI Usage - About once a day	0.39^{***} (0.09)	0.75^{***} (0.12)
AI Usage - Several times a day	0.19 (0.17)	0.78^{***} (0.20)
AI Usage - Almost constantly	-0.31^{**} (0.16)	0.30 (0.25)
AI surgery - Not an advance	-0.16 (0.10)	0.11 (0.17)
AI surgery - A minor advance	0.26^{***} (0.09)	0.71^{***} (0.14)
AI surgery - A major advance	-0.61^{***} (0.17)	-0.89^{***} (0.29)
AI weather - Not an advance	-0.13 (0.10)	-0.15 (0.15)
AI weather - A minor advance	0.12 (0.09)	0.36^{**} (0.14)
AI weather - A major advance	0.50^{***} (0.08)	0.58^{***} (0.16)
AI healthcare - Somewhat uncomfortable	1.12^{***} (0.09)	1.68^{***} (0.15)
AI healthcare - Somewhat comfortable	0.96^{***} (0.17)	2.52^{***} (0.21)
AI healthcare - Very comfortable	0.08 (0.26)	1.23^{***} (0.36)
AI econ impact - Not an advance	0.005 (0.11)	0.65^{***} (0.19)
AI econ impact - A minor advance	-0.34^{***} (0.10)	0.21 (0.18)
AI econ impact - A major advance	-0.14^{**} (0.07)	-0.57^{***} (0.10)
Gender - Female	-0.54 (0.34)	-0.85^{*} (0.50)
Gender - Other	-0.26^{**} (0.13)	-0.23 (0.17)
Age category - 30-49	-0.49^{***} (0.13)	-0.37^{**} (0.18)
Age category - 50-64	-0.44^{***} (0.14)	-0.50^{***} (0.19)
Age category - 65+	0.18^{*} (0.09)	0.03 (0.15)
Education level - Some College	0.06 (0.09)	0.25^{*} (0.14)
Education level - College graduate+	0.05 (0.10)	0.38^{***} (0.14)
Race - Black	0.11 (0.10)	0.51^{***} (0.14)
Race - Hispanic	0.02 (0.18)	0.14 (0.26)
Race - Other	0.59^{***} (0.21)	0.91^{***} (0.24)
Race - Asian	-0.12 (0.32)	-3.31^{***} (0.63)
Akaike Inf. Crit.	9,566.73	9,566.73
Note:		*p<0.1; **p<0.05; ***p<0.01

In the above results, the reference category is "More Concerned than excited" (for perception of AI). For Internet Usage – used almost constantly is the only significant and positive relationship in both groups (column 1 – balanced attitudes towards AI and column 2 – excited attitudes towards AI). This is a new finding compared to the Ordered logit model and definitely makes sense since a vast majority of AI tools are only accessible via the internet.

The AI usage frequency shows interesting patterns. For “Equally concerned and excited” (column 1), using AI “About once a day” positively affects users’ outlook (0.39***), while “Almost constantly” negatively affects it (-0.31**). This suggests moderate users are more balanced than infrequent users on their perception of AI. For “More excited than concerned” (column 2), AI usage frequency positively affects users’ excitement (0.75*** and 0.78***), indicating regular users in this category are more excited than concerned about AI.

For AI surgery as "A minor advance" increases likelihood of balanced (0.26***) or excited (0.71***) attitudes, while viewing it as "A major advance" decreases likelihood of a lower AI perception in both groups.

For AI weather as "A major advance" significantly increases likelihood of balanced (0.50***) or excited (0.58***) attitudes. Interestingly, for AI Economic Impact, "A minor advance" (-0.34***) or "A major advance" (-0.14**) decreases likelihood of balanced attitudes, with similar negative effects for the excited group (-0.57***).

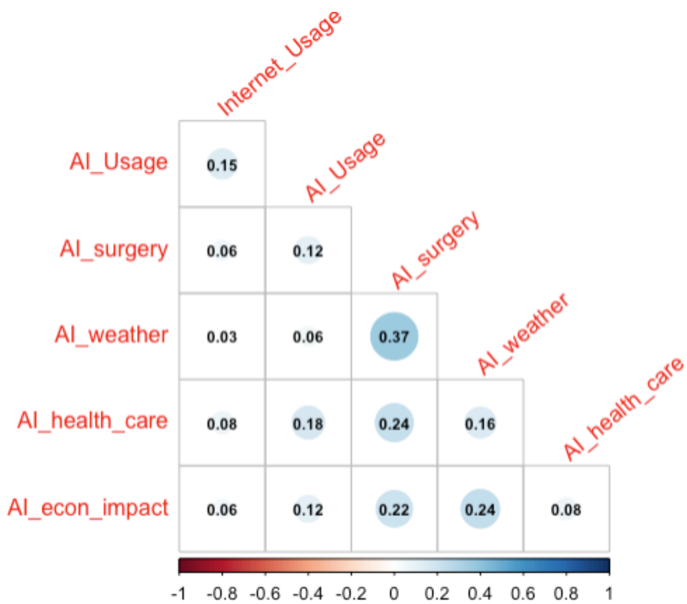
Most of the demographic variables follow a similar pattern as the Ordered Logit model except Race. Here, Asians are less likely to be more excited than concerned about AI which is completely opposite from the Ordered Logit model results. This means that responses from Asians (compared to reference category: White) cannot be captured by a simple ordered relationship. Race as a variable is quite complex and requires a further nuanced set of analysis

The overall pattern indicates that moderate AI usage, positive attitudes toward specific AI applications (especially in healthcare and weather forecasting), and demographic factors significantly influence whether individuals exhibit equal concern or excitement, or greater excitement than concern, regarding AI.

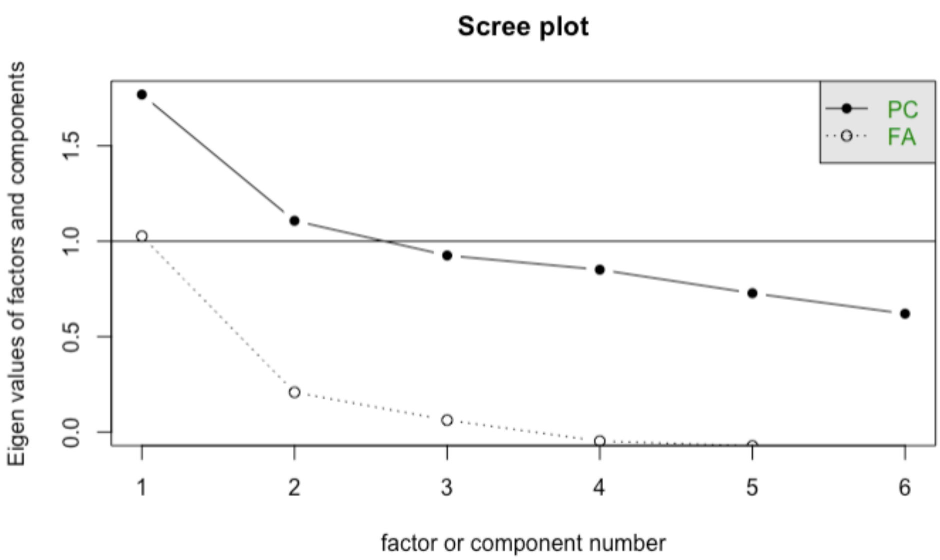
Compared to the Ordered Logit model results, AI use in healthcare also seems to be the most important variable in the Multinomial model results.

Approach 2 - EFA Exploratory Factor Analysis & New Scale Creation

Initial analysis of correlations of the independent variables shows that AI for surgery has low positive correlation with AI for weather forecasting, AI use in healthcare and AI economic impact. This suggests that there could be 2 to 4 possibly distinct factors.



The scree plot below shows that there are possibly 2 factors (eigenvalues >= 1)



Since the factors represent difference in attitudes based on different domains (Health care, Robotic Surgery, Weather forecasting, Economic impact) but attempt to capture perception of AI when AI is used in those domains, I am allowing for the factors to be correlated with each other. Therefore, “oblimin” orthogonal rotation is used along with factoring method “ml” for maximum likelihood.

Results of running EFA – 2 factors

Tucker-Lewis Index (TLI): 0.92
This is slightly below the 0.95 which is indicative of the 2-factor model not being an almost well-fitting model

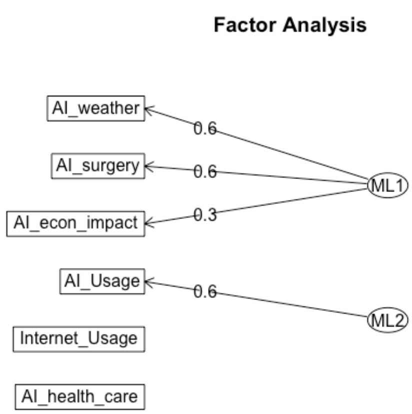
RMSEA: 0.044
This is below the 0.05 level which is indicative of a well-fitting model

ML1 loadings > 0.3
AI Surgery: 0.59
AI Weather: 0.64
AI Economic Impact: 0.34

ML 2 Loadings > 0.3
AI Usage: 0.57

ML 1 Proportion explained: 66%
ML 2 Proportion explained: 34%

90 % confidence intervals are: 0.029 0.06



Results of running EFA – 3 factors

Tucker-Lewis Index (TLI): -Inf

RMSEA: 0

Both of these indicate that a 3-factor model is a not a good model fit

Results of running EFA – 4 factors

Tucker-Lewis Index (TLI): 1.014

While this is above the 0.95 level for a well-fitting model, the df degrees of freedom is negative (-3)

The results above indicate that a 2-factor model is the best fitting. Creating a new measure based on the 3 IV's identified by the 2-factor model: AI Weather, AI Surgery, AI Economic Impact.

After creating a new measure "AI Factors", the alpha analysis shows the following:

Cronbach's Raw Alpha: 0.76
This is above the 0.7 threshold and means that this scale is reliable

	raw_alpha <dbl>
AI_weather	0.7333547
AI_surgery	0.7372096
AI_econ_impact	0.7928513
AI_factors	0.5370834

The raw alpha for the individual items shows that the new measure "AI Factors" makes a substantial positive contribution to reliability. If it were removed, the alpha would drop to 0.53.

If AI Economic impact is removed, then alpha would increase to 0.79, suggesting that this could be removed and a new measure using just AI Weather and AI Surgery could be further analyzed.

The correlation plot below shows that the new measure has high correlations with the underlying variables, but doesn't have perfect correlation. This indicates that the new scale was created successfully

Conclusion

The dependent variable, AI Perception, is ordinal with three categories. An ordered logit model was initially utilized due to the ordinal nature of the outcome. However, the Brant test indicated that the proportional odds assumption was violated. When this assumption does not hold, the ordered logit model may yield biased or misleading results. To address this, a multinomial logistic regression model was used instead, as it does not require the proportional odds assumption and is appropriate for modeling outcomes with more than two categories, regardless of their order.

The quantitative analysis suggests that Comfort with AI in Healthcare is the Strongest Predictor. Both the Order Logit and Multinomial models show that as people become more comfortable with AI in healthcare, their perception of AI shifts dramatically in a positive direction. The coefficients for comfort levels are large and highly significant.

EFA (Exploratory Factor Analysis) shows that combining different variables related to AI usage in different domains can be combined to simplify factors that attempt to capture perception of AI.

Since our world is complex, the quantitative methods provide a toolbox of various models that can be utilized depending on the structure of how we are attempting to model the outcome.

To enhance public perception of artificial intelligence (AI), corporations and governments must prioritize fostering a sense of comfort and familiarity among individuals with AI. This can be achieved through various means, including educational initiatives, transparent communication, and the promotion of positive examples. Comfort and experience serve as significant drivers of enthusiasm and acceptance towards AI.

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